FORM-ADAPTIVE CLAMPING SYSTEMS

ALVERY Clamping Systems

Clamping technology for machining, quality control, assembly and automation

Product range 2025

Alvery Spanntechnik GmbH – fictional company for assessment purposes

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The Alvery system

Self-shaping fixtures with repeatable accuracy

Alvery develops modular clamping systems that adapt to the workpiece instead of the other way round. Where conventional fixtures are designed for one part and stored for years, an Alvery module takes the shape of whatever part is placed into it and can be reset in seconds.

The idea is borrowed from the pin-art toy: hundreds of pins that reproduce any shape pressed into them. Alvery turned this principle into precise, lockable industrial modules. They are used in machining, in measuring rooms, in assembly and, increasingly, on robot grippers and driverless transport systems.

All modules of a line share the same interfaces. Adapters, angles, base plates and profile plates can be combined freely, so a set bought for one task can be extended for the next one. Typical customers report savings of 50 to 80 % in fixture construction and storage.

This catalogue lists the standard sets of each line. Every component of a set can also be ordered individually. Technical values for each module are summarised on the set pages; binding values are given in the module data sheets.

The basic principle

Every Alvery module contains a field of individually movable, spring-loaded pins. The workpiece is pressed into the pin field, the pins follow its contour, and a single clamping movement (screw, hand lever, compressed air or electric actuator) locks all pins at once. The result is a form-fit support that holds the part without point loads.

Releasing the lock returns the pins to their initial position, ready for the next geometry. No part of the module has to be machined or adapted for a new workpiece.

Technical fundamentals

Pin stroke

The pin stroke is the maximum distance a pin can travel. It limits the height difference of the contour that a module can reproduce. As a rule of thumb, the pin stroke should be at least 1.5 times the height difference of the supported contour. If the contour exceeds the stroke, the outer pins reach their end position and the workpiece is only supported on a few pins.

Pin diameter

Fine pins (1 mm and 3 mm) reproduce small details and are used for light parts. 6 mm pins are the standard for machining and metrology. 14 mm pins carry large, heavy parts in automation. Smaller pins mean more pins per area and a finer reproduction of the contour, but a lower load per pin.

Holding force

The axial holding force is the maximum force in pin direction that the locked pin field can absorb without pins moving. For manual PRO modules it depends directly on the tightening torque of the clamping screw; the data sheets of the FormVise modules contain a torque table. Pneumatic modules distinguish between passive holding force (spring and friction, without compressed air) and active holding force (with 6 bar).

Lateral forces are absorbed by the form fit and are not specified separately. For machining, Alvery recommends a safety factor of 2 between the expected cutting force and the axial holding force.

Repeatability

A workpiece placed again into a locked pin field returns to the same position within 0,02 mm, provided the pin field is clean. Chips and dried coolant are the most frequent cause of poor repeatability.

Selecting the right line

Line	Typical use	Pin Ø	Clamping	Typical holding force
METRO	Coordinate measuring machines, optical metrology, CT, assembly	4–6 mm	hand lever	up to 2 kN
PRO	Machining in vises, mould clamping, production	3–6 mm	clamping screw	1,8–9,5 kN
MICRO	Watch parts, medical, electronics	1 mm	clamping screw	low
AUTO	Robot grippers, assembly lines, AGVs	3–14 mm	pneumatic / electric	95 N – 4,2 kN

Start from the process, not from the product: in a measuring room the workpiece must not be deformed, so low clamping forces and good accessibility matter most. In machining, cutting forces dominate and the holding force of the module is the limiting value. In automation, cycle time and the actuation type (pneumatic or electric) decide.

For workpieces heavier than about 10 kg on a single gripper, combine several modules or use the FormBed series. For CT scans, only the CT variants with carbon pins and foam tips should be used, because steel pins cause artefacts in the scan.

METRO LINE

METRO LINE – Metrology and assembly

METRO LINE sets are designed for measuring rooms and assembly. The modules are made of anodized aluminium with stainless steel pins and are locked with a hand lever, so set-up does not need tools.

Measuring complex parts such as turbine blades, cast housings or trim parts requires a support that holds the part securely without deforming it and without blocking the probe. METRO modules combine form-fit support with very low clamping forces.

Overview of METRO LINE sets

Order no.	Set	Modules	Pin Ø	Pin stroke
42.3168	FormRest XL Set	4 x FormRest XL module	6 mm	22 mm
42.3169	FormRest XL Light Set	2 x FormRest XL module	6 mm	22 mm
42.1210	FormRest SXS Set	6 x FormRest SXS module	6 mm	12 mm
42.1424	FormRest XS Set	4 x FormRest XS module	4 mm	10 mm
42.6225	FormGrip XL Set	2 x FormGrip XL module	6 mm	22 mm
42.5815	FormGrip XS Set	4 x FormGrip XS module	3 mm	10 mm
42.5813	FormGrip XS Light Set	2 x FormGrip XS module	3 mm	10 mm
48.2550	ProfileRest System	6 x FormRest XL module	6 mm	22 mm
47.1001	ShiftRail Set	2 x FormRest XL module	6 mm	22 mm
47.1200	BaseTable 1200	accessory set	–	–
49.0510	GridPlate Set	accessory set	–	–

FORMREST XL SET

FormRest XL Set

ORDER NUMBER: 42.3168

The workhorse for coordinate measuring machines. Four FormRest XL modules on a large T-slot plate hold cast housings, turbine blades and sheet metal parts without deformation.

The set is built around 4 FormRest XL modules with 72 pins of 6 mm diameter and 22 mm stroke, locked by hand lever.

Delivery: complete in an Alvery case with foam inlay.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	42.0670	4	FormRest XL module	52 x 105 x 95 mm	32 x 82 mm	72 pcs	6 mm	22 mm
			Further information about the module: Dimension 52 x 105 x 95 mm, pin field 32 x 82 mm, weight: 1,5 kg, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
2	42.6727	4	FormRest connecting element, base plate	50 x 100 mm	–	–	–	–
3	42.6728	4	FormRest base plate	50 x 100 mm	–	–	–	–
4	42.3120	4	Horizontal clamp	M8, clamping force 1,2 kN	–	–	–	–
5	42.3125	4	Pressure pad	Ø 20 mm, PU	–	–	–	–
6	42.3130	2	Cone support	Ø 20 mm, height 40 mm	–	–	–	–
7	42.3131	2	Prism support 90°	40 x 40 mm	–	–	–	–
8	42.3140	4	Height extension	50 mm	–	–	–	–
9	49.0400	1	T-slot profile plate	400 x 300 mm	–	–	–	–
10	49.0501	2	Clamping brackets for T-slot plate (1 set = 3 pcs)	–	–	–	–	–
11	49.2501	2	Fastening set (1 set = 8 pcs)	–	–	–	–	–
12	49.9000	1	Alvery case	–	–	–	–	–

FORMREST XL LIGHT SET

FormRest XL Light Set

ORDER NUMBER: 42.3169

The smaller FormRest XL set for compact measuring machines.

The set is built around 2 FormRest XL modules with 72 pins of 6 mm diameter and 22 mm stroke, locked by hand lever.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	42.0670	2	FormRest XL module	52 x 105 x 95 mm	32 x 82 mm	72 pcs	6 mm	22 mm
			Further information about the module: Dimension 52 x 105 x 95 mm, pin field 32 x 82 mm, weight: 1,5 kg, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
2	42.6727	2	FormRest connecting element, base plate	50 x 100 mm	–	–	–	–
3	42.6728	2	FormRest base plate	50 x 100 mm	–	–	–	–
4	42.3120	2	Horizontal clamp	M8, clamping force 1,2 kN	–	–	–	–
5	42.3125	2	Pressure pad	Ø 20 mm, PU	–	–	–	–
6	49.0200	1	T-slot profile plate	250 x 200 mm	–	–	–	–
7	49.2501	1	Fastening set (1 set = 8 pcs)	–	–	–	–	–

FORMREST SXS SET

FormRest SXS Set

ORDER NUMBER: 42.1210

Six compact supports for long and flat parts such as profiles and panels.

The set is built around 6 FormRest SXS modules with 34 pins of 6 mm diameter and 12 mm stroke, locked by hand lever.

Delivery: complete in an Alvery case with foam inlay.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	42.0310	6	FormRest SXS module	52 x 63 x 80 mm	32 x 39 mm	34 pcs	6 mm	12 mm
			Further information about the module: Dimension 52 x 63 x 80 mm, pin field 32 x 39 mm, weight: 0,8 kg, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
2	42.3140	6	Height extension	50 mm	–	–	–	–
3	42.3130	4	Cone support	Ø 20 mm, height 40 mm	–	–	–	–
4	49.0400	1	T-slot profile plate	400 x 300 mm	–	–	–	–
5	49.2501	2	Fastening set (1 set = 8 pcs)	–	–	–	–	–
6	49.9000	1	Alvery case	–	–	–	–	–

FORMREST XS SET

FormRest XS Set

ORDER NUMBER: 42.1424

Small supports with 4 mm pins for light parts on optical measuring machines.

The set is built around 4 FormRest XS modules with 15 pins of 4 mm diameter and 10 mm stroke, locked by hand lever.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	42.0150	4	FormRest XS module	40 x 40 x 70 mm	22 x 25 mm	15 pcs	4 mm	10 mm
			Further information about the module: Dimension 40 x 40 x 70 mm, pin field 22 x 25 mm, weight: 0,4 kg, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
2	42.6728	4	FormRest base plate	50 x 100 mm	–	–	–	–
3	49.0200	1	T-slot profile plate	250 x 200 mm	–	–	–	–
4	49.2501	1	Fastening set (1 set = 8 pcs)	–	–	–	–	–

FORMGRIP XL SET

FormGrip XL Set

ORDER NUMBER: 42.6225

Gripping set with two lever-operated FormGrip XL modules that clamp the workpiece from both sides.

The set is built around 2 FormGrip XL modules with 72 pins of 6 mm diameter and 22 mm stroke, locked by hand lever.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	42.0675	2	FormGrip XL module	52 x 120 x 110 mm	32 x 82 mm	72 pcs	6 mm	22 mm
			Further information about the module: Dimension 52 x 120 x 110 mm, pin field 32 x 82 mm, weight: 2,1 kg, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
2	42.3120	2	Horizontal clamp	M8, clamping force 1,2 kN	–	–	–	–
3	49.0400	1	T-slot profile plate	400 x 300 mm	–	–	–	–
4	49.0501	2	Clamping brackets for T-slot plate (1 set = 3 pcs)	–	–	–	–	–
5	49.2501	1	Fastening set (1 set = 8 pcs)	–	–	–	–	–

FORMGRIP XS SET

FormGrip XS Set

ORDER NUMBER: 42.5815

Four fine-pin grippers for small parts in assembly and measurement.

The set is built around 4 FormGrip XS modules with 60 pins of 3 mm diameter and 10 mm stroke, locked by hand lever.

Delivery: complete in an Alvery case with foam inlay.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	42.0580	4	FormGrip XS module	34 x 73 x 60 mm	22 x 25 mm	60 pcs	3 mm	10 mm
			Further information about the module: Dimension 34 x 73 x 60 mm, pin field 22 x 25 mm, weight: 0,45 kg, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
2	42.6728	4	FormRest base plate	50 x 100 mm	–	–	–	–
3	49.0200	1	T-slot profile plate	250 x 200 mm	–	–	–	–
4	49.2501	1	Fastening set (1 set = 8 pcs)	–	–	–	–	–
5	49.9000	1	Alvery case	–	–	–	–	–

FORMGRIP XS LIGHT SET

FormGrip XS Light Set

ORDER NUMBER: 42.5813

Two fine-pin grippers, the entry into the FormGrip XS system.

The set is built around 2 FormGrip XS modules with 60 pins of 3 mm diameter and 10 mm stroke, locked by hand lever.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	42.0580	2	FormGrip XS module	34 x 73 x 60 mm	22 x 25 mm	60 pcs	3 mm	10 mm
			Further information about the module: Dimension 34 x 73 x 60 mm, pin field 22 x 25 mm, weight: 0,45 kg, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
2	42.6728	2	FormRest base plate	50 x 100 mm	–	–	–	–
3	49.2501	1	Fastening set (1 set = 8 pcs)	–	–	–	–	–

PROFILEREST SYSTEM

ProfileRest System

ORDER NUMBER: 48.2550

Large modular system for long profiles, frames and car body parts. Supports can be positioned anywhere along the aluminium profiles.

The set is built around 6 FormRest XL modules with 72 pins of 6 mm diameter and 22 mm stroke, locked by hand lever.

Delivery: complete in an Alvery case with foam inlay.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	42.0670	6	FormRest XL module	52 x 105 x 95 mm	32 x 82 mm	72 pcs	6 mm	22 mm
			Further information about the module: Dimension 52 x 105 x 95 mm, pin field 32 x 82 mm, weight: 1,5 kg, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
2	42.0310	4	FormRest SXS module	52 x 63 x 80 mm	32 x 39 mm	34 pcs	6 mm	12 mm
3	42.6727	6	FormRest connecting element, base plate	50 x 100 mm	–	–	–	–
4	42.6728	10	FormRest base plate	50 x 100 mm	–	–	–	–
5	42.3120	6	Horizontal clamp	M8, clamping force 1,2 kN	–	–	–	–
6	42.3125	6	Pressure pad	Ø 20 mm, PU	–	–	–	–
7	42.3130	4	Cone support	Ø 20 mm, height 40 mm	–	–	–	–
8	42.3131	4	Prism support 90°	40 x 40 mm	–	–	–	–
9	42.3140	8	Height extension	50 mm	–	–	–	–
10	47.0101	2	ShiftRail linear rail	500 mm	–	–	–	–
11	47.0102	6	ShiftRail carriage	50 x 100 mm, lockable	–	–	–	–
12	49.0400	2	T-slot profile plate	400 x 300 mm	–	–	–	–
13	49.0501	4	Clamping brackets for T-slot plate (1 set = 3 pcs)	–	–	–	–	–
14	49.2501	6	Fastening set (1 set = 8 pcs)	–	–	–	–	–
15	43.0050	8	Connector pin for MICRO modules	Ø 5 mm	–	–	–	–
16	41.0042	8	Connection pin for FormVise	M10 x 42 mm	–	–	–	–
17	41.0027	16	Connection pin for FormVise	M10 x 27 mm	–	–	–	–
18	42.3110	4	Fixing elbow for hold-down clamps	–	–	–	–	–
19	42.3112	2	Hold-down set (carbon fiber)	3 x 120 mm, 2 x 250 mm	–	–	–	–
20	49.0200	2	T-slot profile plate	250 x 200 mm	–	–	–	–
21	–	1	Torque wrench	6–30 Nm	–	–	–	–
22	49.9000	2	Alvery case	–	–	–	–	–

SHIFTRAIL SET

ShiftRail Set

ORDER NUMBER: 47.1001

Two linear rails with lockable carriages for quick repositioning of supports.

The set is built around 2 FormRest XL modules with 72 pins of 6 mm diameter and 22 mm stroke, locked by hand lever.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	47.0101	2	ShiftRail linear rail	500 mm	–	–	–	–
2	47.0102	4	ShiftRail carriage	50 x 100 mm, lockable	–	–	–	–
3	42.0670	2	FormRest XL module	52 x 105 x 95 mm	32 x 82 mm	72 pcs	6 mm	22 mm
			Further information about the module: Dimension 52 x 105 x 95 mm, pin field 32 x 82 mm, weight: 1,5 kg, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
4	42.6727	2	FormRest connecting element, base plate	50 x 100 mm	–	–	–	–
5	49.2501	2	Fastening set (1 set = 8 pcs)	–	–	–	–	–

BASETABLE 1200

BaseTable 1200

ORDER NUMBER: 47.1200

Mobile clamping table for large set-ups next to the measuring machine.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	47.1210	1	BaseTable top, aluminium, T-slot	1200 x 800 x 40 mm	–	–	–	–
2	47.1220	1	BaseTable frame	1200 x 800 x 750 mm	–	–	–	–
3	47.1230	4	Caster with brake	Ø 100 mm	–	–	–	–
4	49.0400	2	T-slot profile plate	400 x 300 mm	–	–	–	–

GRIDPLATE SET

GridPlate Set

ORDER NUMBER: 49.0510

Bore-grid plate for repeatable positioning of cone supports.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	49.0510	1	GridPlate, bore grid 25 mm	300 x 300 x 15 mm	–	–	–	–
2	42.3130	4	Cone support	Ø 20 mm, height 40 mm	–	–	–	–
3	49.2501	1	Fastening set (1 set = 8 pcs)	–	–	–	–	–

MICRO LINE

MICRO LINE – Delicate and small workpieces

MICRO LINE modules use 1 mm pins and reproduce even very small contours. They are made for watch parts, medical implants, connectors and other delicate parts that are measured or assembled by hand.

MICRO modules are connected with Ø 5 mm connector pins and can be combined into small fixtures on a base rail or clamped into a standard vise via the vise adapter.

Overview of MICRO LINE sets

Order no.	Set	Modules	Pin Ø	Pin stroke
43.2515	FormGrip Micro S	2 x FormGrip Micro Single module	1 mm	8 mm
43.2520	FormGrip Micro M	2 x FormGrip Micro Single module	1 mm	8 mm
43.2530	FormGrip Micro L	4 x FormGrip Micro Single module	1 mm	8 mm

FORMGRIP MICRO S

FormGrip Micro S

ORDER NUMBER: 43.2515

The smallest set: two 1 mm pin modules for watch parts, implants and electronics.

Typical applications: watch parts; connectors; implants.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	43.0126	2	FormGrip Micro Single module	21 x 21 x 52 mm	15 x 12 mm	130 pcs	1 mm	8 mm
			Further information about the module: Dimension 21 x 21 x 52 mm, pin field 15 x 12 mm, weight: 70 g, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
2	43.0050	4	Connector pin for MICRO modules	Ø 5 mm	–	–	–	–
3	43.0100	1	MICRO base rail	200 mm	–	–	–	–

FORMGRIP MICRO M

FormGrip Micro M

ORDER NUMBER: 43.2520

Single and double modules on a base rail, with angle brackets for vertical set-ups.

Typical applications: watch parts; connectors; implants.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	43.0126	2	FormGrip Micro Single module	21 x 21 x 52 mm	15 x 12 mm	130 pcs	1 mm	8 mm
			Further information about the module: Dimension 21 x 21 x 52 mm, pin field 15 x 12 mm, weight: 70 g, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
2	43.0252	1	FormGrip Micro Double module	21 x 42 x 52 mm	15 x 12 (2x) mm	260 pcs	1 mm	8 mm
3	43.0050	6	Connector pin for MICRO modules	Ø 5 mm	–	–	–	–
4	43.0100	1	MICRO base rail	200 mm	–	–	–	–
5	43.0110	2	MICRO angle bracket	40 x 40 mm	–	–	–	–

FORMGRIP MICRO L

FormGrip Micro L

ORDER NUMBER: 43.2530

The complete MICRO set, including a vise adapter for use in standard 80 mm vises.

Typical applications: watch parts; connectors; implants.

Delivery: complete in an Alvery case with foam inlay.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	43.0126	4	FormGrip Micro Single module	21 x 21 x 52 mm	15 x 12 mm	130 pcs	1 mm	8 mm
			Further information about the module: Dimension 21 x 21 x 52 mm, pin field 15 x 12 mm, weight: 70 g, housing aluminium (anodized), pins stainless steel, installation position: any, cleaning: compressed air, data sheet and CAD data on request					
2	43.0252	2	FormGrip Micro Double module	21 x 42 x 52 mm	15 x 12 (2x) mm	260 pcs	1 mm	8 mm
3	43.0050	12	Connector pin for MICRO modules	Ø 5 mm	–	–	–	–
4	43.0100	2	MICRO base rail	200 mm	–	–	–	–
5	43.0110	4	MICRO angle bracket	40 x 40 mm	–	–	–	–
6	43.0120	1	MICRO vise adapter	for vises with 80 mm jaws	–	–	–	–
7	49.9000	1	Alvery case	–	–	–	–	–

PRO LINE – Production, assembly and machining

PRO LINE modules are made for production: machining, mould clamping and heavy assembly. Housings and pins are stainless steel, and the pin field is locked with a clamping screw and a torque wrench.

FormVise modules replace soft jaws in standard machine vises. The pins follow the workpiece contour, and after locking the module works like a jaw that was milled for exactly this part. Always tighten the clamping screw with the torque wrench to the value given in the data sheet; the holding force depends directly on it.

Overview of PRO LINE sets

Order no.	Set	Modules	Pin Ø	Pin stroke
41.1070	FormVise 100 Base Kit Small	3 x FormVise 100 (symmetric) module	6 mm	22 mm
41.1071	FormVise 100 Base Kit Large	4 x FormVise 100 (symmetric) module	6 mm	22 mm
41.1068	Accessories: FormVise 100 Jaw Set	2 x FormVise 100 (symmetric) module	6 mm	22 mm
41.1286	Accessories: FormVise 125 Jaw Set	2 x FormVise 125 (symmetric) module	6 mm	22 mm
41.1128	Accessories: FormVise 127W Jaw Set	2 x FormVise 127W (symmetric) module	6 mm	22 mm
41.3859	FormVise 40 Mini Base Kit Small	1 x FormVise 40 Mini module	3 mm	12 mm
41.3860	FormVise 40 Mini Base Kit Large	2 x FormVise 40 Mini module	3 mm	12 mm
41.3139	Accessories: FormRest 50	2 x FormRest 50 module	6 mm	12 mm

FORMVISE 100 BASE KIT SMALL

FormVise 100 Base Kit Small

ORDER NUMBER: 41.1070

The entry kit for mould clamping in a standard machine vise. Three FormVise 100 modules work as form jaws or as pin supports, so the same vise holds cast parts, forgings and prototypes without dedicated jaws.

Typical applications: mould clamping of cast and forged parts in machine vises; prototype machining; small batches with changing geometries.

Clamping: tighten each module with 25 Nm using the torque wrench. Axial holding force per module: 7,5 kN.

Delivery: complete in an Alvery case with foam inlay.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	41.0672	3	FormVise 100 (symmetric) module	52 x 120 x 100 mm	32 x 82 mm	72 pcs	6 mm	22 mm
			Further information about the module: Dimension 52 x 110 x 100 mm, pin field 32 x 83 mm, weight: 3,3 kg, material housing/pins: stainless steel					
2	41.1026	2	Adapter plate for FormVise 100	100 x 100 x 18,5 mm	–	–	–	–
3	41.1039	2	Adapter angle (fastening bores optional)	100 x 100 x 50 mm	–	–	–	–
4	41.1015	2	Adapter for FormVise 100	50 x 100 x 15 mm	–	–	–	–
5	41.1016	2	Adapter for FormVise 100	50 x 10 x 30 mm	–	–	–	–
6	41.1030	1	Angle for FormVise 100	30°, 50 x 100 mm	–	–	–	–
7	41.1031	1	Angle for FormVise 100	30°, 100 x 50 mm	–	–	–	–
8	41.0120	1	S-base plate for FormVise 100	130 x 100 x 10 mm	–	–	–	–
9	41.0100	2	Base plate for FormVise	50 x 150 x 10 mm	–	–	–	–
10	41.0042	4	Connection pin for FormVise	M10 x 42 mm	–	–	–	–
11	41.0027	22	Connection pin for FormVise	M10 x 27 mm	–	–	–	–
12	–	1	Torque wrench	6–30 Nm	–	–	–	–
13	49.9000	1	Alvery case	–	–	–	–	–

FORMVISE 100 BASE KIT LARGE

FormVise 100 Base Kit Large
ORDER NUMBER: 41.1071

The complete kit for workshops that clamp many different part families. Four modules plus a full set of adapters and base plates allow horizontal, vertical and angled set-ups.

Typical applications: mould clamping of cast and forged parts in machine vises; prototype machining; small batches with changing geometries.

Clamping: tighten each module with 25 Nm using the torque wrench. Axial holding force per module: 7,5 kN.

Delivery: complete in an Alvery case with foam inlay.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	41.0672	4	FormVise 100 (symmetric) module	52 x 120 x 100 mm	32 x 82 mm	72 pcs	6 mm	22 mm
			Further information about the module: Dimension 52 x 110 x 100 mm, pin field 32 x 83 mm, weight: 3,3 kg, material housing/pins: stainless steel					
2	41.1026	4	Adapter plate for FormVise 100	100 x 100 x 18,5 mm	–	–	–	–
3	41.1039	2	Adapter angle (fastening bores optional)	100 x 100 x 50 mm	–	–	–	–
4	41.1015	4	Adapter for FormVise 100	50 x 100 x 15 mm	–	–	–	–
5	41.1016	4	Adapter for FormVise 100	50 x 10 x 30 mm	–	–	–	–
6	41.1030	2	Angle for FormVise 100	30°, 50 x 100 mm	–	–	–	–
7	41.1031	2	Angle for FormVise 100	30°, 100 x 50 mm	–	–	–	–
8	41.0120	2	S-base plate for FormVise 100	130 x 100 x 10 mm	–	–	–	–
9	41.0100	4	Base plate for FormVise	50 x 150 x 10 mm	–	–	–	–
10	41.0042	8	Connection pin for FormVise	M10 x 42 mm	–	–	–	–
11	41.0027	30	Connection pin for FormVise	M10 x 27 mm	–	–	–	–
12	–	1	Torque wrench	6–30 Nm	–	–	–	–
13	49.9000	2	Alvery case	–	–	–	–	–

ACCESSORIES: FORMVISE 100 JAW SET

Accessories: FormVise 100 Jaw Set

ORDER NUMBER: 41.1068

Two FormVise 100 modules with adapter plates, mounted directly on the jaws of a machine vise.

Typical applications: mould clamping of cast and forged parts in machine vises; prototype machining; small batches with changing geometries.

Clamping: tighten each module with 25 Nm using the torque wrench. Axial holding force per module: 7,5 kN.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	41.0672	2	FormVise 100 (symmetric) module	52 x 120 x 100 mm	32 x 82 mm	72 pcs	6 mm	22 mm
			Further information about the module: Dimension 52 x 120 x 100 mm, pin field 32 x 82 mm, weight: 3,2 kg, material housing/pins: stainless steel					
2	41.1026	2	Adapter plate for FormVise 100	100 x 100 x 18,5 mm	–	–	–	–

ACCESSORIES: FORMVISE 125 JAW SET

Accessories: FormVise 125 Jaw Set

ORDER NUMBER: 41.1286

Jaw set for vises with 125 mm jaw width. Higher holding force for heavier workpieces.

Typical applications: heavier workpieces in 125 mm vises; 5-axis machining of housings.

Clamping: tighten each module with 25 Nm using the torque wrench. Axial holding force per module: 9,0 kN.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	41.0852	2	FormVise 125 (symmetric) module	52 x 145 x 100 mm	32 x 104 mm	90 pcs	6 mm	22 mm
			Further information about the module: Dimension 52 x 145 x 100 mm, pin field 32 x 104 mm, weight: 4,0 kg, material housing/pins: stainless steel					
2	41.1226	2	Adapter plate for FormVise 125	125 x 100 x 18,5 mm	–	–	–	–

ACCESSORIES: FORMVISE 127W JAW SET

Accessories: FormVise 127W Jaw Set

ORDER NUMBER: 41.1128

Wide pin field for large contact areas. Recommended for thin-walled housings.

Typical applications: thin-walled housings; large contact areas, low surface pressure.

Clamping: tighten each module with 30 Nm using the torque wrench. Axial holding force per module: 9,5 kN.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	41.1272	2	FormVise 127W (symmetric) module	78 x 128 x 100 mm	56 x 82 mm	132 pcs	6 mm	22 mm
			Further information about the module: Dimension 78 x 128 x 100 mm, pin field 56 x 82 mm, weight: 4,8 kg, material housing/pins: stainless steel					
2	41.1227	2	Adapter plate for FormVise 127W	128 x 100 x 20 mm	–	–	–	–

FORMVISE 40 MINI BASE KIT SMALL

FormVise 40 Mini Base Kit Small

ORDER NUMBER: 41.3859

Miniature machine vise with two 3 mm pin fields for small and delicate parts in production.

Typical applications: small precision parts; medical and dental parts; watch components.

Clamping: tighten each module with 12 Nm using the torque wrench. Axial holding force per module: 1,8 kN.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	41.0386	1	FormVise 40 Mini module	50 x 265 x 78 mm	22 x 25 (2x) mm	60 (2x) pcs	3 mm	12 mm
			Further information about the module: Dimension 50 x 265 x 78 mm, pin field 22 x 25 (2x) mm, weight: 3,2 kg, material housing/pins: stainless steel					
2	41.3870	2	Jaw adapter for FormVise 40 Mini	50 x 48 x 12 mm	–	–	–	–
3	41.3880	1	Base plate for FormVise 40 Mini	50 x 250 x 10 mm	–	–	–	–
4	–	1	Torque wrench	2–12 Nm	–	–	–	–

FORMVISE 40 MINI BASE KIT LARGE

FormVise 40 Mini Base Kit Large

ORDER NUMBER: 41.3860

Two miniature vises for parallel machining of small parts.

Typical applications: small precision parts; medical and dental parts; watch components.

Clamping: tighten each module with 12 Nm using the torque wrench. Axial holding force per module: 1,8 kN.

Delivery: complete in an Alvery case with foam inlay.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	41.0386	2	FormVise 40 Mini module	50 x 265 x 78 mm	22 x 25 (2x) mm	60 (2x) pcs	3 mm	12 mm
			Further information about the module: Dimension 50 x 265 x 78 mm, pin field 22 x 25 (2x) mm, weight: 3,2 kg, material housing/pins: stainless steel					
2	41.3870	4	Jaw adapter for FormVise 40 Mini	50 x 48 x 12 mm	–	–	–	–
3	41.3880	2	Base plate for FormVise 40 Mini	50 x 250 x 10 mm	–	–	–	–
4	–	1	Torque wrench	2–12 Nm	–	–	–	–
5	49.9000	1	Alvery case	–	–	–	–	–

ACCESSORIES: FORMREST 50

Accessories: FormRest 50

ORDER NUMBER: 41.3139

Compact pin supports for long workpieces, combinable with all FormVise base plates.

Typical applications: support of long workpieces in machining; counter-support against vibration.

Clamping: tighten each module with 25 Nm using the torque wrench. Axial holding force per module: 3,7 kN.

Delivery: in a cardboard box; the Alvery case (49.9000) is available as an option.

The modular system consists of the following individual components:

Fig.	Article no.	Qty	Description	Dimension	Pin field	Number of pins	Pin Ø	Pin stroke
1	41.0312	2	FormRest 50 module	52 x 63 x 72 mm	32 x 39 mm	34 pcs	6 mm	12 mm
			Further information about the module: Dimension 52 x 63 x 72 mm, pin field 32 x 39 mm, weight: 1,25 kg, material housing/pins: stainless steel					
2	41.0100	2	Base plate for FormVise	50 x 150 x 10 mm	–	–	–	–
3	41.0027	8	Connection pin for FormVise	M10 x 27 mm	–	–	–	–

AUTO LINE

AUTO LINE – Automation

AUTO LINE modules bring the pin principle to robots, assembly lines and driverless transport systems. They are actuated pneumatically (AutoForm, PickForm, FormBed p) or electrically (FormBed e), or locked by hand on manual stations (FormBed m).

AUTO modules are sold individually and are not part of the sets in this catalogue.

Order no.	Module	Clamping	Pins	Pin Ø	Stroke	Holding force
44.0672	AutoForm 67	pneumatic	72	6 mm	22 mm	4,2 kN active
44.0580	AutoForm Mini	pneumatic	60	3 mm	10 mm	95 N active
44.0312	PickForm 50	pneumatic	34	6 mm	22 mm	750 N active
44.0142	PickForm 50 F (3 mm pins)	pneumatic	150	3 mm	16 mm	see data sheet
45.0830	FormBed m	manual	88	14 mm	32 mm	2,1 kN
45.0832	FormBed p	pneumatic	88	14 mm	32 mm	2,1 kN
45.0831	FormBed e	electric	88	14 mm	32 mm	2,1 kN

Detailed information on the AUTO LINE is given in the AutoForm and FormBed flyers and in the module data sheets.

Equipment and accessories

All accessories can be ordered separately and are compatible with every Alvery line, unless stated otherwise.

Article no.	Description	Dimension
41.0027	Connection pin for FormVise	M10 x 27 mm
41.0042	Connection pin for FormVise	M10 x 42 mm
41.0100	Base plate for FormVise	50 x 150 x 10 mm
41.0120	S-base plate for FormVise 100	130 x 100 x 10 mm
41.1015	Adapter for FormVise 100	50 x 100 x 15 mm
41.1016	Adapter for FormVise 100	50 x 10 x 30 mm
41.1026	Adapter plate for FormVise 100	100 x 100 x 18,5 mm
41.1030	Angle for FormVise 100	30°, 50 x 100 mm
41.1031	Angle for FormVise 100	30°, 100 x 50 mm
41.1039	Adapter angle (fastening bores optional)	100 x 100 x 50 mm
41.1226	Adapter plate for FormVise 125	125 x 100 x 18,5 mm
41.1227	Adapter plate for FormVise 127W	128 x 100 x 20 mm
41.3870	Jaw adapter for FormVise 40 Mini	50 x 48 x 12 mm
41.3880	Base plate for FormVise 40 Mini	50 x 250 x 10 mm
42.3110	Fixing elbow for hold-down clamps	–
42.3111	Hold-down set (carbon fiber)	3 x 120 mm, 2 x 300 mm
42.3112	Hold-down set (carbon fiber)	3 x 120 mm, 2 x 250 mm
42.3120	Horizontal clamp	M8, clamping force 1,2 kN
42.3125	Pressure pad	Ø 20 mm, PU
42.3130	Cone support	Ø 20 mm, height 40 mm
42.3131	Prism support 90°	40 x 40 mm
42.3140	Height extension	50 mm
42.6701	Adapter for hold-down clamps	–
42.6727	FormRest connecting element, base plate	50 x 100 mm
42.6728	FormRest base plate	50 x 100 mm
42.6757	Carbon pin extension with PE foam tip (incl. module 42.0672)	72 pins, Ø 6 mm, stroke 22 mm
43.0050	Connector pin for MICRO modules	Ø 5 mm
43.0100	MICRO base rail	200 mm
43.0110	MICRO angle bracket	40 x 40 mm
43.0120	MICRO vise adapter	for vises with 80 mm jaws
47.0101	ShiftRail linear rail	500 mm
47.0102	ShiftRail carriage	50 x 100 mm, lockable
47.1210	BaseTable top, aluminium, T-slot	1200 x 800 x 40 mm
47.1220	BaseTable frame	1200 x 800 x 750 mm
47.1230	Caster with brake	Ø 100 mm
49.0200	T-slot profile plate	250 x 200 mm
49.0400	T-slot profile plate	400 x 300 mm
49.0501	Clamping brackets for T-slot plate (1 set = 3 pcs)	–
49.0510	GridPlate, bore grid 25 mm	300 x 300 x 15 mm
49.2501	Fastening set (1 set = 8 pcs)	–
49.9000	Alvery case	–

Alvery interfaces

Alvery modules are compatible with common third-party systems. PRO modules fit standard machine vises with 100 mm and 125 mm jaw width via adapter plates. METRO modules use 50 x 100 mm base plates that can be mounted on any T-slot plate with M8 slot nuts.

AUTO modules are mounted with 4 x M5 screws and can be integrated into robot flanges (ISO 9409-1-50-4-M6 via adapter) and AGV platforms. Pneumatic modules need 6 bar filtered compressed air; electrically locked modules need 24 V DC.

Pin variants

Variant	Pin Ø	Tip	Material	Used in
Standard	6 mm	flat	stainless steel	FormVise, FormRest, FormGrip XL, AutoForm 67, PickForm 50
Fine	3 mm	flat	stainless steel	FormVise 40 Mini, FormGrip XS, AutoForm Mini, PickForm 50 F
Micro	1 mm	rounded	stainless steel	FormGrip Micro
Heavy	14 mm	flat	hardened steel	FormBed m / p / e
CT	6 mm / 8 mm tip	PE foam	carbon fiber	FormRest XL 100 CT, FormBed CT
Soft	6 mm	POM cap	stainless steel, POM	on request for all 6 mm modules

Special pin lengths and tip geometries are available on request. Soft caps reduce the holding force by about 15 %.

Application reports

Gearbox housings on coordinate measuring machines

A supplier of aluminium gearbox housings measures 14 housing variants on two coordinate measuring machines. Each variant had its own fixture. A change-over took up to 40 minutes, and fixtures filled two shelving units.

Solution: One FormRest XL Set (42.3168) per machine. Housings are placed on four pin supports and locked by hand.

- Change-over time reduced from 40 minutes to under 3 minutes
- Repeatability within 0,02 mm over 30 placements
- Payback within 5 months

Turbine blade roots in 5-axis machining

A manufacturer of turbine components machines blade roots of 9 blade types in small batches. Soft jaws were milled for each blade type; every new type required two days of jaw design and machining.

Solution: FormVise 140 H modules (41.0674) as vise jaws. The 42 mm pin stroke covers the twisted blade profile.

- No more soft jaws for new blade types
- Set-up time per batch reduced by 70 %
- Clamping torque 25 Nm, holding force 7,5 kN per module, sufficient for roughing

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Subject to technical changes. Weights and dimensions are approximate values. Binding specifications are given in the current data sheet of each module.